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Embedding Geometric Heuristics for Improving RAG Systems

הטמעת יוריסטיקות גיאומטריות לשיפור מערכות RAG

Research proposal

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Abstract

Retrieval-Augmented Generation (RAG) is the standard way to let a large language model answer questions about documents it was never trained on. A component called the retriever first finds a few documents that seem relevant to the question, and the language model then writes an answer based on them. If the retriever brings the wrong documents, the answer is wrong, no matter how good the language model is. Today's retrievers turn every question and every document into a point in a high-dimensional space and return the documents whose points are closest to the question's point. This works only if a question really does land close to the documents that answer it. That assumption is rarely examined, and this research examines it directly.

The research has three goals. The first is to study the geometric properties of the space in which a retriever places questions and documents: where a question lands relative to the documents that answer it, and what distinguishes the questions on which the retriever succeeds from those on which it fails. The second is to develop simple geometric heuristics, built on what the first goal discovers, that improve retrieval by moving the starting point of the search away from the question and toward the region where the relevant documents are. The third is to develop iterative search algorithms for the cases in which the retriever fails: algorithms that ask a judge about documents one at a time and keep moving the starting point as evidence accumulates. The three goals are evaluated on a benchmark collection of biomedical questions in which every question-document pair has already been scored for relevance by a language model acting as a judge, so that any method can be compared against the standard retriever by simulation, without new judgments, and the results are validated on two public collections of questions and documents.

תקציר

RAG היא הדרך המקובלת לאפשר למודל שפה גדול לענות על שאלות לגבי מסמכים שהוא מעולם לא אומן עליהם. רכיב שנקרא מנגנון השליפה מוצא תחילה כמה מסמכים שנראים רלוונטיים לשאלה, ומודל השפה כותב תשובה על בסיסם. אם השליפה מביאה את המסמכים הלא נכונים, התשובה שגויה, לא משנה כמה טוב מודל השפה. מנגנוני השליפה של היום הופכים כל שאלה וכל מסמך לנקודה במרחב רב-ממדי, ומחזירים את המסמכים שהנקודות שלהם הקרובות ביותר לנקודה של השאלה. זה עובד רק אם השאלה אכן נוחתת קרוב למסמכים שעונים עליה. ההנחה הזו כמעט לא נבדקת, והמחקר הזה בודק אותה ישירות.

למחקר שלוש מטרות. הראשונה היא לחקור את התכונות הגיאומטריות של המרחב שבו מנגנון השליפה ממקם שאלות ומסמכים: היכן שאלה נוחתת ביחס למסמכים שעונים עליה, ומה מבדיל בין השאלות שבהן השליפה מצליחה לאלה שבהן היא נכשלת. השנייה היא לפתח יוריסטיקות גיאומטריות פשוטות, שנשענות על מה שהמטרה הראשונה מגלה, שמשפרות את השליפה על ידי הזזת נקודת ההתחלה של החיפוש מהשאלה לכיוון האזור שבו נמצאים המסמכים הרלוונטיים. השלישית היא לפתח אלגוריתמי חיפוש איטרטיביים למקרים שבהם השליפה נכשלת: אלגוריתמים ששואלים שופט על מסמך אחר מסמך וממשיכים להזיז את נקודת ההתחלה ככל שמצטברות ראיות. שלוש המטרות מוערכות על אוסף מבחן של שאלות ביו-רפואיות שבו כל זוג שאלה-מסמך כבר קיבל ציון רלוונטיות ממודל שפה ששימש כשופט, כך שכל שיטה ניתנת להשוואה מול השליפה הסטנדרטית בסימולציה, בלי שיפוטים חדשים, והתוצאות מאומתות על שני אוספים ציבוריים של שאלות ומסמכים.

Terms used in this proposal

Term (* = ours)	Meaning in this proposal
Embedding, embedding space	An embedding is a list of numbers (a vector) representing a text; similar meanings give similar vectors. The embedding space is the space (768 dimensions here) in which each question and document is a point.
Normalized vector	A vector scaled to length 1. All embeddings here are normalized, so only direction matters and cosine similarity is a plain dot product, which is cheaper to compute.
Cosine similarity / distance	The standard way to compare two embeddings: the cosine of the angle between them (1 = same direction, 0 = unrelated). Distance is $1 - \text{similarity}$, so smaller means closer.
Retriever, bi-encoder	The retriever returns the documents it finds most relevant to a question. In a bi-encoder, one network embeds the question, one embeds each document, and the two are compared afterwards.
Nearest-neighbor search, pool	Returning the documents whose embeddings are closest to a given point. The pool is the set of documents searched.
Search anchor*	The point from which nearest-neighbor search is run: the question's embedding in a standard retriever. Moving it is the central idea of this proposal.
Heuristic	A simple rule of thumb that usually works well but is not guaranteed to be the best.
Rank, Top-k	The position of a document in a method's output list (1 = first); the top-k are the k documents ranked highest.
Ground truth (GT)	The one document the benchmark marks as the correct answer to a question.
Relevance judgment, judge	A score for how relevant a document is to a question, here given by an LLM acting as a judge, from 0.1 (irrelevant) to 1.0 (fully relevant).
Exact / soft success*	Exact: a method reaches the ground-truth document. Soft: it reaches any document the judge scores 1.0.
Centroid	The average of a set of embeddings, scaled back to length 1: the "center of mass" of a group of documents.
Oracle centroid*	An oracle uses information a real system would not have at run time, to measure what is possible. The oracle centroid is the centroid of all documents the judge scored 1.0 for a question: the center of the truly relevant region.
Inspection, budget*	To inspect a document is to ask the judge for its score; the budget is how many inspections a method may make.

1. Scientific background

This section builds up the problem in four steps. It first explains what a RAG system is and why its retriever decides the quality of the final answer. It then describes how retrievers are built today and introduces the idea of a *search anchor*, which is the object this research works on. It then explains why retrievers are not perfect, using what is known about the geometry of embedding spaces. Finally it presents our approach and what has been done so far.

1.1 What is RAG

A large language model (LLM) answers questions from what it absorbed during training. That knowledge is frozen on the day training ends and does not include private or specialized documents. Retrieval-Augmented Generation (RAG) fixes this by adding a search step in front of the model (Lewis et al., 2020). A RAG system has three components:

1. An **indexer**, which runs once in advance: it turns every document in a collection into an embedding (a vector of numbers that captures the document's meaning) and stores all the vectors.
2. A **retriever**, which runs for every question: it turns the question into an embedding in the same way, finds the documents whose embeddings are closest to it, and returns the top few.
3. A **generator**, the LLM itself: it receives the question together with the retrieved documents and writes the answer from them.

Many variations of this pipeline exist. Some feed many retrieved passages to the generator at once (Izacard and Grave, 2021), some let the model decide when it needs to retrieve (Jiang et al., 2023; Asai et al., 2024), and a recent survey organizes the whole family (Gao et al., 2023b). What all variations share is the dependence on the retriever: if the document that holds the answer is not among the retrieved ones, the generator cannot produce a correct grounded answer, and documents that are irrelevant actively push it toward wrong answers (Shi et al., 2023). The quality of the retriever is therefore an upper bound on the quality of the whole system, and it is the retriever that this research studies.

How the retriever decides what is close. The standard retriever is a *bi-encoder* (Karpukhin et al., 2020): the same kind of neural network encodes questions and docu-

ments separately into vectors of the same length (768 numbers in our case). Each vector is scaled to length 1, so that only its direction carries meaning, and the retriever ranks documents by the *cosine similarity* between the question's vector and each document's vector. Cosine similarity is the cosine of the angle between two vectors: 1 when they point in the same direction, 0 when they are unrelated. Returning the documents with the highest similarity is called *nearest-neighbor search*. The whole design rests on one assumption: that a question's vector lands close to the vectors of the documents that answer it.

1.2 How retrievers are implemented today

Overview of approaches. Research on retrievers has improved every part of the design except that assumption. One line trains better encoders, so that similar meanings map to closer vectors: the network is trained to pull matching question-document pairs together and push other pairs apart, on unlabeled text (Izacard et al., 2022), on web-scale weak labels (Wang et al., 2022), with instructions attached to the text (Su et al., 2023), and with carefully chosen "hard negatives", documents that are close to the question but wrong (Qu et al., 2021; Xiong et al., 2021). A second line scores documents in a richer way than one cosine between two vectors: one vector per word (Khattab and Zaharia, 2020), or vectors with one entry per vocabulary word (Formal et al., 2021). A third line adds a *re-ranker*: a slower, stronger model, often an LLM, that re-orders the top candidates the retriever returned (Nogueira et al., 2020; Sun et al., 2023). A fourth line rewrites the question before encoding it, for example by asking an LLM to write a short hypothetical answer and searching with that text instead; the best-known version is HyDE, short for Hypothetical Document Embeddings (Wang et al., 2023; Gao et al., 2023a). A fifth line, the oldest, is *relevance feedback*: take the top-ranked documents (the top-k), assume they are relevant, and move the question's vector toward their average; this is known as the Rocchio update (Rocchio, 1971), and it works on embeddings as well as on words (Li et al., 2023). A recent survey covers all of these (Zhao et al., 2024).

What is a search anchor. In every method above, the search is run from some point in the embedding space, and the documents returned are the nearest neighbors of that point. We call that point the *search anchor*, a term we introduce; the literature speaks only of the query representation. In a standard retriever, the anchor is the question's embedding. Rewriting methods replace it by the embedding of the rewritten text. Relevance feedback

moves it toward the top-ranked documents. A few recent methods move it using feedback from an LLM: TOUR (Test-time Optimization of qUery Representations) adjusts the anchor step by step, using a re-ranker's opinions as a stand-in for relevance labels (Sung et al., 2023); ReDE-RF (Real Document Embeddings from Relevance Feedback) asks an LLM which first-pass documents are relevant and searches from their embeddings instead (Jedidi et al., 2024); BAGEL, a Bayesian active-learning method, spends a limited number of LLM judgments to explore the space (Kim et al., 2026). Re-rankers, by contrast, never move the anchor: they only re-order what the anchor already found. The distinction matters, because a document that the anchor misses stays missed no matter how good the re-ranker is.

1.3 Gap: why retrievers are not perfect

Even the best retrievers rank the correct document first for only part of the questions, and the failures are not random. Two kinds of knowledge about the embedding space help explain why.

What is known about the space. Three facts about embedding spaces are well established. First, the vectors do not spread out evenly. They crowd into a narrow cone, so even two unrelated texts look fairly similar, and the differences that matter are squeezed into a small range (Ethayarajh, 2019). Second, in a space with many dimensions some documents become "hubs". A hub is a document that shows up among the nearest neighbors of many different questions, not because it answers them but because of where it happens to sit in the space (Radovanović et al., 2010). Third, one vector per text can only tell so many documents apart (Luan et al., 2021). The usual fixes change all vectors in the same way, for example by removing the directions that every vector shares (Mu and Viswanath, 2018), and such fixes do help retrievers (Jung et al., 2023). There is also evidence that questions and documents end up in different parts of the space. A bi-encoder never sees a question and a document together. Its training pulls questions toward documents but not the other way around (Li et al., 2021), and whether the two encoders share their weights changes where each group lands (Dong et al., 2022).

What is not known. All of this describes the space as a whole. What has not been measured, to our knowledge, is the situation of one question at a time: where the question's point sits relative to the documents that are relevant to it, and relative to the one

document marked as the correct answer. If a question systematically lands *away* from its relevant documents, then nearest-neighbor search from the question's point is searching in the wrong place, and no re-ranker can fix that.

How relevance is measured. Studying this requires knowing, for many question-document pairs, how relevant the document is. Human labels do not exist at that scale, so we use an LLM as a *judge*: it reads a question and a document and returns a score between 0.1 and 1.0. LLM judges agree with human assessors about as well as assessors agree with each other (Thomas et al., 2024; Upadhyay et al., 2024), although they have known biases and should not be treated as the final word (Clarke and Dietz, 2025). In this proposal the judge is a measuring instrument: it tells us where the relevant documents are, so that the geometry can be measured. Its scores also reveal something that a single "correct answer" label hides: most questions have several documents that answer them equally well. We therefore count success in two ways throughout, reaching the labeled answer (*exact success*) and reaching any document the judge rates as fully relevant (*soft success*); both terms are ours.

1.4 Our approach

Our approach is to measure first and design second. The first goal is to study the geometry of the retriever's space one question at a time: how crowded the question's neighborhood is, how far it is from the documents that answer it, and which of these properties separate the questions the retriever solves from those it fails. The intention is a picture of the space that is precise enough to act on, and, as its practical output, a model that predicts from geometry alone whether the retriever will fail on a given question. The second goal takes what the first discovers and turns it into geometric heuristics: simple rules that move the search anchor from the question toward the region where the relevant documents lie, by an amount set from the measured geometry rather than fixed in advance. The intention is a better starting point for the search that any bi-encoder can use without retraining. The third goal addresses the cases where the retriever fails even so: iterative search algorithms that inspect documents one at a time, ask the judge about each, and move the anchor again after each answer, so that the search follows the evidence it collects. The intention is to reach a relevant document with as few inspections as possible exactly where the standard retriever needs the most help.

Each goal already has a first result behind it. For the first, we measured the space on a benchmark of 196 biomedical questions with a 5,035-document pool in which every pair is judged. The question's neighborhood holds 3 documents on average while the center of the relevant documents is surrounded by 10, that center is 44% closer to the correct answer than the question is, and the questions on which the retriever fails differ from the others in these measurements. For the second, we compared the standard retriever with several fixed ways of blending the question with that center: a small pull toward it gains one point of accuracy, replacing the question by that center loses most of the accuracy, and 79% of the retriever's "failures" return a document the judge rates as fully relevant. For the third, a first prototype that moved the anchor by a fixed amount after every relevant hit made results worse, from 3.3 to 4.6 inspected documents on average; the analysis of why gives the design of the proposed algorithm.

2. Research objectives and expected significance

The three goals correspond to three research questions.

RQ1. What does the retriever's embedding space look like around a question, and does that geometry tell us when the retriever will fail? Concretely: where does a question's point sit relative to its relevant documents and its correct answer, which measurable properties of the neighborhood separate solved questions from failed ones, and can a model built from those properties alone predict failure before any document is inspected?

Expected contribution: a measured description of the space and a failure predictor that needs no relevance labels at run time.

RQ2. Can a geometric heuristic that moves the search anchor improve retrieval? If the anchor is moved from the question toward the relevant region, by an amount chosen from the geometry measured for RQ1, does the retriever find the correct document, or a fully relevant one, after inspecting fewer documents than the standard retriever?

Expected contribution: a one-shot anchor heuristic and the conditions under which it helps.

RQ3. For the questions on which the retriever fails, does an iterative search that keeps moving the anchor do better? If the anchor is updated after ev-

ery inspected document, using the judge's answer, does the search reach a relevant document faster than the one-shot heuristic and than existing feedback methods, at the same inspection budget?

Expected contribution: an iterative algorithm, its cost-accuracy curve, and a comparison with the closest published methods.

Significance. Retrieval quality bounds every RAG answer, yet the standard retriever rests on an assumption, that a question lands near its answers, that is not checked. Answering RQ1 gives a way to know in advance when that assumption fails for a given question. Answering RQ2 and RQ3 gives a way to act on it without retraining the encoder or re-indexing the collection: the heuristics operate on the vectors any bi-encoder already produces, so they can be added to an existing system. The biomedical benchmark is the place where the methods are compared, chosen because the gap between questions and documents is known to be large there; the findings are then validated on two public collections with different questions and documents, so that the conclusions are about retrievers in general and not about one dataset.

3. Detailed description of the proposed research

The research is organized in four phases. Phase 1 is the shared foundation: the benchmark on which everything is measured, the baselines against which everything is compared, and the measurements that motivate the rest; most of it is complete. Phases 2, 3 and 4 correspond to RQ1, RQ2 and RQ3. Each is described in the same way: what problem it solves, stated precisely; how we propose to solve it; and how the solution will be evaluated.

Notation. Before the phases, a few symbols that all of them share. A question is represented by its embedding, which we write Q; a document by its embedding, d; and the document the benchmark marks as the correct answer by GT. All embeddings are normalized to length 1, so the cosine similarity between two of them, written $\mathrm{sim}(x, y)$, is simply their dot product, and the cosine distance $d(x, y) = 1 - \mathrm{sim}(x, y)$ is 0 for identical vectors and grows as they point in different directions. For scale, a typical document in the benchmark pool lies at distance about 0.9 from a question and the nearest one at about 0.15, so a distance such as 0.17 is small. We write $\mathrm{normalize}(v)$ for the vector v scaled to length 1. The pool of documents is $\mathcal{D}$. The judge's relevance score for a document is $J(d)$, a number between 0.1 and 1.0; we say that we *inspect* a document when we obtain this

score, and the *inspection budget* is the number of documents a method may inspect (both terms are ours). The set of documents the judge scores 1.0, the fully relevant ones, is R, and C is the centroid of R: the average of their vectors, scaled back to length 1. We call C the **oracle centroid**: the center of the region of documents that are truly relevant to this question, found by using the judge's score for every document in the pool. A running system cannot compute it, because finding the relevant documents is exactly the problem it is trying to solve. We therefore use C only as a yardstick, to measure where the relevant region is and what a search could achieve if it knew that location. Later, when a method builds a centroid from only the documents it has inspected so far, we write that centroid with a subscript, C_k or C_t, to make clear that it is not the oracle.

3.1 Phase 1. Setup and benchmark

The benchmark. Comparing retrieval methods requires a fixed set of questions, a fixed set of documents, and, for each question, a way to tell how good every method's output is. The collection we use for this is a biomedical question-answering benchmark derived from PubMed, the main bibliographic database of the life sciences: 196 questions, each with one abstract marked as its correct answer, and a shared pool of 5,035 abstracts. We chose a biomedical collection as the benchmark, rather than a general one, because this is where the gap between questions and documents is known to be largest: retrievers trained on general text lose the most on biomedical tasks (Thakur et al., 2021), the questions are short expert queries while the documents are long technical abstracts (Tsatsaronis et al., 2015), and pre-training on biomedical text alone does not close the gap (Lee et al., 2020; Jin et al., 2023). A large gap makes the geometry easier to measure and a method's improvement easier to see. The benchmark is where the methods are compared; the domain itself is not the object of the research, and Phases 2-4 each end with a validation on two public collections, TREC-COVID and MS MARCO.

Every question and every document in the benchmark was embedded with the same publicly available bi-encoder (`all-mpnet-base-v2`, 768 dimensions). Then every question-document pair, about one million pairs, was scored for relevance by a Gemini-2.5-flash-lite judge; the scores fall almost entirely on six values (0.1, 0.2, 0.4, 0.6, 0.8, 1.0). This exhaustive judging is what makes the rest of the research practical. A retrieval method is, in the end, a rule for the order in which documents are inspected, and since the benchmark already holds the judge's answer for every document, any method can be simulated offline,

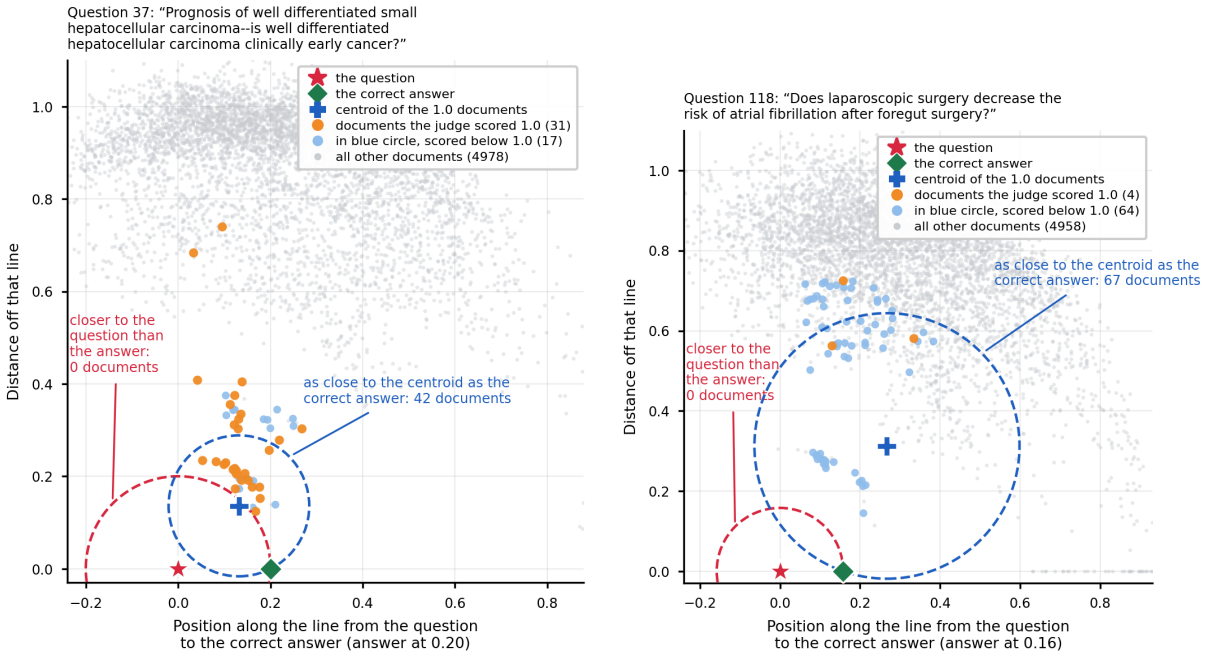
on all 196 questions, in seconds, with no new LLM calls. The judging also showed that the correct answer is rarely unique: on average 17 documents per question receive the top score.

Baseline retrievers. A new method is only as convincing as the methods it is compared against, so Phase 1 fixes a set of baselines that spans the design space of Section 1.2 and can all be simulated on the benchmark. The first and most important is **B1**, the standard retriever, which ranks the pool by distance to the question, $d(d, Q)$; every other method must beat it. The second, **B2**, ranks the pool by distance to the oracle centroid, $d(d, C)$; it is not a deployable method, but it shows what searching from the center of the relevant region would give if that center were known. Between these two we placed three fixed blends, **H1-H3**: a weighted sum of the two distances, $\alpha\, d(d, Q) + (1 - \alpha)\, d(d, C)$ for several values of α; the question's order with the centroid breaking ties; and B1's top-k re-ordered by distance to the centroid. These map out what a rule fixed in advance can do, and they are the reference point for the geometry-guided rules of Phases 3 and 4. Two baselines represent the feedback family: classic **relevance feedback**, in which B1's top-k are judged and the anchor is moved to the centroid of those judged relevant (Li et al., 2023), and **LLM re-ranking**, in which B1's top-k are re-ordered by the judge's score (Sun et al., 2023). All of these are implemented. The remaining baselines are the published anchor-moving methods, ReDE-RF (Jedidi et al., 2024) and TOUR (Sung et al., 2023), and the hypothetical-document anchor of HyDE (Gao et al., 2023a); implementing them on the benchmark is the open item of Phase 1.

What the benchmark shows. Table 1 gives the main numbers and Figure 1 shows the geometry of two questions. Three results drive the rest of the proposal.

Table 1: Where the correct document ends up in each method's output, over 196 questions. "Mean rank" is the average position of the correct document (lower is better); "Top-k" is the share of questions where it is among the first k returned (higher is better); "-" means not measured. Methods marked † use the oracle centroid, which a real system cannot compute.

Method	Mean rank	Top-1 (%)	Top-3 (%)	Top-10 (%)
B1: distance to the question (standard retriever)	3.29	70.4	83.2	91.8
B2†: distance to the oracle centroid	9.87	14.8	36.2	62.8
H1†: $0.93\,d(d,Q) + 0.07\,d(d,C)$	3.35	71.4	82.7	90.8
H2†: B1 order, centroid breaks ties	3.29	70.4	83.2	91.8
H3†: B1's top-10 re-ordered by centroid	5.05	17.9	43.9	91.8
Relevance feedback (top-5 judged, those scored 1.0 kept)	4.90	31.1	-	-
LLM re-ranking of B1's top-5	2.35	40.8	-	-



(a) Question 37. No document is closer to the question than the correct answer is, so the standard retriever ranks the answer first; but 42 documents are as close to the centroid as the correct answer is.

(b) Question 118. Only four documents were scored 1.0, and their centroid lies farther from the correct answer than the question is; its neighborhood is nevertheless crowded.

Figure 1: Two questions drawn in two dimensions; all distances are cosine distances. Each dot is one document. The question (red star) sits at 0 on the horizontal axis and the correct answer (green diamond) at its true cosine distance from the question; every document is then placed so that its distance to the question and its distance to the correct answer, measured in the picture, equal its true cosine distances in the embedding space. The horizontal axis is therefore the position along the line from the question to the correct answer, and the vertical axis is how far a document lies off that line; neither axis is used by any retriever, they are only drawing coordinates. Distances to any other point, including the oracle centroid C (blue cross), are only approximate in this picture. The standard retriever ranks documents by cosine distance to the question, so the documents it returns before the correct answer are exactly those closer to the question than the correct answer is. These form a circle of radius $d(Q, GT)$ around the question (the red circle; a ball in the full space). The blue circle is the same construction around C, with radius $d(C, GT)$. Orange dots are documents the judge scored 1.0; light-blue dots are documents inside the blue circle with a lower score.

Result 1: a question lies away from the center of its relevant documents. For each question we draw a circle around it whose radius is its distance to the correct answer; every document inside that circle is returned before the correct answer. The circle holds 3.1 documents on average (median 1). The same circle drawn around the centroid C holds 9.7 (median 6), while a circle of the same radius around a random point at the same distance from the correct answer holds only 0.8, so the density around C is real and not an artifact of the radius. The mean distances are $d(Q, GT) \approx d(Q, C) \approx 0.17$ and $d(C, GT) \approx 0.10$: the centroid is 44% closer to the correct answer than the question is. The correct answer is not on the straight line between Q and C either; its mean distance

from that line, measured at a right angle, is 0.39. The implication is that neither the question nor the centroid is, by itself, the right place to search from.

Result 2: the judge's score tracks the distance to the correct answer. Documents the judge scores 0.1 lie at mean distance 0.89 from the correct answer, and the distance falls step by step with the score: 0.84, 0.77, 0.63, 0.40, and 0.22 for score 1.0. The relation holds within every one of the 196 questions. The implication is twofold. The judge is a reliable instrument: when it says "fully relevant", the document is geometrically near the answer. But the judge cannot replace geometry: re-ordering B1's top-5 purely by the judge's score lowers Top-1 from 70.4% to 40.8%, because many documents share the top score and the labeled answer is only one of them.

Result 3: fixed rules barely help, and most failures still return a relevant document. In Table 1, a small fixed pull toward the centroid (H1) gains one point of Top-1 over B1; stronger pulls, tie-breaking, and re-ordering a window all lose, and replacing the question by a feedback centroid loses badly (31.1%). Among the 58 questions where B1 does not rank the correct answer first, in 46 (79%) the document it returns first is itself scored 1.0 by the judge, and every one of the 58 has a fully relevant document in its top 10. The implication is that a rule fixed in advance is not enough, and that exact and soft success must be reported side by side, as they are throughout this proposal.

3.2 Phase 2. Characterizing the geometry and predicting retriever failure

Problem definition. Phase 1 measured the geometry in aggregate. Phase 2 turns it into a per-question description and tests whether that description predicts failure. For each question q, let $y_q = 1$ if the standard retriever B1 does not rank the correct document first (an exact failure), and $y_q = 0$ otherwise; on the benchmark, $y_q = 1$ for 58 of the 196 questions. Let $\mathbf{x}_q$ be a vector of features computed only from the question's embedding and the embeddings of the documents in the pool, with no relevance labels and no knowledge of the correct answer, so that a running system could compute it. The task is to learn a function $f(\mathbf{x}_q)$ that estimates the probability that $y_q = 1$. The quality measure is the area under the ROC curve (AUC) of f on questions it was not trained on; AUC is the standard measure of how well a score separates two groups (0.5 is chance, 1.0 is perfect).

Proposed approach. The features describe the neighborhood of the question in the same terms as the measurements of Phase 1, but without using the correct answer. The

first group measures how *far* the question is from the documents around it: the distance to its nearest document, to its 5th nearest, and to its 20th nearest. The second group measures how *crowded* that neighborhood is: how many documents lie within a fixed radius, and how tightly the top-20 cluster around their own center (their average distance to that center). The third group measures how *coherent* the neighborhood is: the distance from the question to the center of its top-20, and the average similarity between pairs of the top-20 documents. The intuition, drawn from Phase 1, is that a failing question tends to sit in a neighborhood that is both distant and ambiguous: the correct answer lies inside the question's circle for 89% of the questions B1 solves but only 74% of those it fails, and failing questions have more fully relevant documents around them (7.5 against 6.5 in the top 10), so several documents compete for the first position. We will fit two standard classifiers, logistic regression and gradient-boosted decision trees (a model that combines many small decision trees), and estimate their accuracy by cross-validation: the 196 questions are split into folds, each fold is predicted by a model trained on the others, and the AUC is computed over all held-out predictions. We will also report which features carry the signal, since the description of the space is itself a result of RQ1.

Evaluation. Cross-validated AUC on the 196 questions, compared against two reference points: a predictor that uses no features, and the coherence-based predictors proposed for bi-encoder retrievers (Vlachou and Macdonald, 2024), which are the closest existing method and predict query difficulty from the spread of the top-ranked embeddings. The same predictor is then trained and tested on two public collections with human relevance labels, TREC-COVID (Roberts et al., 2020), a search task over COVID-19 research papers, and MS MARCO (Nguyen et al., 2016), a collection of web-search questions, to check that the geometric signal is not specific to the biomedical benchmark. The predictor also serves Phases 3 and 4 directly: it tells them which questions to act on and which to leave to the standard retriever.

3.3 Phase 3. A geometry-guided search anchor heuristic

Problem definition. Consider a retriever that may inspect a small number k of documents, meaning send them to the judge, before returning its final ranking. The standard retriever simply inspects B1's top-k and returns B1's order. A *one-shot anchor heuristic* does three things instead: it inspects B1's top-k and obtains their judgments; it computes a new anchor A from the question, the inspected documents, and the geometry around

them; and it ranks the whole pool by $d(d, A)$. The task is to choose the rule that produces A so that the correct document, or a fully relevant one, appears as early as possible in the final ranking. Two existing methods are instances of this scheme. The Rocchio update sets A by moving Q a fixed fraction λ of the way toward C_k, the centroid of the inspected documents the judge found relevant: $A = \text{normalize}\big(Q + \lambda\,(C_k - Q)\big)$. HyDE sets A to the embedding of a hypothetical answer written by an LLM, without inspecting anything.

Proposed approach. The measurements of Phase 1 say what a good rule must respect. A small move toward the relevant region helps and a full move hurts, so λ must stay well below 1. The correct answer lies off the line from Q to the centroid, so the move should be cautious when the inspected evidence is thin. And the retriever already solves 70% of the questions, so the rule must not disturb those. We therefore set λ per question from two measured quantities rather than fixing it. The first is the *support*: how many of the k inspected documents were judged relevant. A centroid built from one document is a noisy estimate of where the relevant region is; one built from five is not; so λ grows with the support. The second is the *spread*: how far the relevant inspected documents are from their own centroid. A tight cluster points clearly in one direction and justifies a larger step; a loose one does not. As a concrete example, with $k = 5$, three documents judged relevant and tightly clustered, the rule moves the anchor roughly a third of the way toward their centroid; with one relevant document, it barely moves at all. On top of this, the failure predictor of Phase 2 acts as a gate: questions it predicts B1 will solve keep $A = Q$ and are never touched. Two variants will be tested alongside: using the HyDE embedding instead of Q as the starting point of the move, and a "repulsion" variant that also pushes A away from the centroid of inspected documents judged *irrelevant*.

Evaluation. All variants are simulated on the benchmark at budgets of 1, 2, 3, 5 and 10 inspections. The measures are the mean number of documents inspected until the correct document is reached (exact success) and until the first fully relevant one is reached (soft success), the Top-1 and Top-10 rates at each budget, and the full cost-accuracy curve (the success rate as a function of the number of inspections). Comparisons are against B1, the fixed blends H1-H3, Rocchio relevance feedback, HyDE, and ReDE-RF, all at the same budget. Differences are tested with a paired bootstrap over questions (a statistical test that resamples the questions many times to check that a difference is not due to chance), and every result is reported separately for the questions B1 solves and the questions it

fails, since Phase 1 shows that the two groups behave differently. The best variant is then re-run on TREC-COVID and MS MARCO.

3.4 Phase 4. An iterative search algorithm for the failing cases

Problem definition. Phase 3 moves the anchor once, after k inspections. For the questions on which the retriever fails, one move may not be enough: the first inspected documents may all be irrelevant, or relevant but scattered, so that no single new anchor is clearly right. The iterative version therefore moves the anchor after *every* inspection. At step t it chooses the next document to inspect, receives the judge's score, updates the anchor, and chooses again, until it reaches a fully relevant document or exhausts a budget of B inspections. Formally, the method is a policy that maps the history of inspected documents and their scores to the next document to inspect, and the task is to minimize the expected number of inspections until success. The published methods closest to this are the iterative feedback loops surveyed by Rathee et al. (2026) and the budgeted exploration of BAGEL (Kim et al., 2026).

Proposed approach. It helps to first follow one question through the algorithm in words. The search starts exactly like B1: the anchor is the question, and the first document inspected is the one nearest to it. If the judge scores that document 1.0, the search is over and nothing was lost. If not, the algorithm now knows that the question's own neighborhood is misleading, and from this point on the anchor is allowed to move. Every document the judge scores at least θ joins a set R of relevant documents. Once R holds at least m documents, the anchor is moved to a point between the question and the centroid of R,

$$A = \text{normalize}\big(a\,Q + (1-a)\,\text{centroid}(R)\big), \qquad a = \alpha(|R|), \tag{1}$$

where the weight a starts near 1 (stay close to the question) and decreases as R grows, so that the anchor slides from the question toward the relevant region at a pace set by the evidence. Each new document to inspect is simply the uninspected document nearest to the current anchor. The search stops at the first document scored 1.0, or when the budget of B inspections runs out. Algorithm 1 states this precisely. Two thresholds appear in it on purpose: θ decides which documents count as evidence for moving the anchor, and the score 1.0 decides when the search has succeeded.

Algorithm 1 Iterative geometry-guided anchoring (Phase 4)

Require: question Q, pool $\mathcal{D}$, judge J, budget B; parameters θ, m, schedule $\alpha(\cdot)$

$A \leftarrow Q$ ▷ anchor: start from the question, exactly like B1
$R \leftarrow \emptyset$; move $\leftarrow$ false
for $t = 1, \ldots, B$ **do**
  $d \leftarrow$ the uninspected document nearest to A
  $s \leftarrow J(d)$ ▷ one inspection
  if $s = 1.0$ **then return** d after t inspections ▷ success
  end if
  if $s \geq \theta$ **then** add d to R ▷ evidence
  else move $\leftarrow$ true ▷ a miss unlocks movement
  end if
  if move **and** $|R| \geq m$ **then**
    $a \leftarrow \alpha(|R|)$ ▷ less weight on Q as R grows
    $A \leftarrow \text{normalize}\big(a\,Q + (1-a)\,\text{centroid}(R)\big)$ ▷ move the anchor
  end if
end for
return failure after B inspections

The design follows directly from the failed prototype described in Section 1.4. That prototype used a fixed weight, began moving the anchor after the first relevant hit, and rebuilt the centroid from whatever it had; its average number of inspections rose from 3.29 (B1) to 4.55 as the weight on the question decreased. Each of its flaws becomes a design decision here: (i) the anchor starts moving only after an inspected document is judged *not* relevant, so questions that B1 solves immediately are never disturbed; (ii) it moves only once the centroid rests on at least m relevant documents; and (iii) the weight is a schedule that decreases with the evidence rather than a constant. In the offline simulation the loop is also allowed to continue past its first success, so that the step at which it reaches the labeled answer (exact success) is recorded in the same run as the step at which it reaches a fully relevant document (soft success).

Evaluation. The same measures and the same offline simulation as Phase 3, now as a function of the budget B (1, 2, 3, 5 and 10 inspections): mean inspections to exact and to soft success, success rates at each budget, and the cost-accuracy curve. Comparisons are

against B1, the one-shot heuristic of Phase 3 (to isolate the value of iterating), iterative Rocchio feedback, TOUR, and BAGEL, all under the same budget. Because every judgment is pre-computed, one setting of Algorithm 1 runs on all 196 questions in seconds, so the parameters θ, m and the schedule can be swept exhaustively and the sweep itself reported, including the settings that do *not* work. As in the previous phases, results are split by whether B1 solves the question, which is where this phase is expected to matter, and the best variant is validated on TREC-COVID and MS MARCO. Where those collections need graded judgments to simulate the loop, the volume (a few hundred questions against pools of a few thousand documents) is of the same order as the one million pairs already judged for the benchmark.

3.5 Time schedule

The work is planned over three semesters, from October 2026 to March 2028. Phase 1 is largely done; its remaining item is implementing the published anchor-moving baselines. The phases overlap because all of them run on the same offline benchmark and none waits for new data.

Table 2: Gantt chart. S1: Oct 2026-Mar 2027; S2: Apr-Sep 2027; S3: Oct 2027-Mar 2028.

	S1	S2	S3
Phase 1: remaining baseline retrievers			
Phase 2: geometry and failure prediction, validation			
Phase 3: geometry-guided anchor heuristic			
Phase 4: iterative search algorithm, validation			
Paper writing and submission			
Thesis writing			

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